



Second Phase Presentation

PROJECT AND TEAM INFORMATION

Project Title

Intelligent Digital Asset Management System

Student / Team Information

<i>Team Name:</i>	<i>Semicolon Haters</i>
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PROPOSAL DESCRIPTION

Project Abstract

The main goal of this project is to develop an **Intelligent Digital Asset Management System (IDAMS)** that goes beyond simple file storage and enables smart organization, retrieval, and management of digital assets. Traditional systems depend on manual tagging, folder structures, and exact filenames, which often leads to inefficiency, duplication, and difficulty in locating relevant files. This project aims to solve these limitations by integrating Artificial Intelligence into the asset management process.

The system is designed to automatically understand the content of uploaded assets such as images, videos, and documents. It generates meaningful tags, detects duplicate or similar files, and creates vector embeddings to support semantic search. This allows users to search for assets using natural language queries and retrieve results based on meaning rather than exact keywords.

Additionally, the project focuses on maintaining proper version control, ensuring that users can track and access different versions of an asset. It also incorporates role-based access control to enhance security and manage user permissions effectively.

Overall, the goal is to transform a conventional storage system into an intelligent, efficient, and scalable platform that reduces manual effort, improves search accuracy, prevents redundancy, and enhances collaboration among users.

Updated Project Approach and Architecture

Our system follows a **modular layered architecture** with clearly defined components: **React frontend → Flask REST API → AI service layer → PostgreSQL database → FAISS vector index → local file storage.**

The frontend is built using React (with hooks) and styled using Tailwind CSS, providing an interactive dashboard for users. It communicates with the backend through **REST APIs over HTTP/HTTPS**, using Axios to send and receive data in JSON format, along with support for multipart file uploads. Authentication is handled using **JWT-based authentication**, where tokens are issued during login/registration and included in requests via the Authorization

header. Role-Based Access Control (RBAC) is enforced at the backend using role claims.

The backend is implemented using Flask, managing core business logic, authentication, and API endpoints. It uses SQLAlchemy for database interaction and PostgreSQL to store metadata such as asset details, tags, user data, and version history. Key functionalities include asset upload, filtering, version tracking, semantic search, and secure downloads. During upload, files are hashed to detect exact duplicates, stored in a local file system, and metadata is recorded in the database.

The AI layer is responsible for adding intelligence to the system. It uses PyTorch and transformer-based models to generate embeddings for images and text. These embeddings are stored in a FAISS index, enabling fast similarity-based search. Automatic tagging is performed using similarity matching for visual data and zero-shot classification for textual content.

The system is configurable through environment variables, allowing easy adaptation across different environments while maintaining scalability and flexibility.

Tasks Completed

Tasks Completed	Team Member
The system architecture and backend setup have been established. Basic RESTful APIs for user registration, login, and asset upload have been implemented.	Aditya Sharma
The frontend dashboard has been initiated using React and Tailwind CSS. Basic pages like login, upload, and asset listing are available.	Ayush Bisht
The AI/ML pipeline is in progress. Initial work on embedding generation has started, but full semantic search integration using FAISS is not yet complete.	Kushagra Garg
The database design using PostgreSQL has been created, and core tables for users and assets are functional.	Apratim Rana

Challenges/Roadblocks

During the development of the Intelligent Digital Asset Management System, we have faced several key challenges.

One major issue has been **integrating the AI pipeline with the backend**, as generating embeddings and tags can slow down the upload process. We plan to solve this by shifting AI tasks to **asynchronous background processing**.

Another challenge is **improving semantic search accuracy**, since initial results are sometimes not very relevant. We aim to refine this by tuning models and adjusting similarity thresholds in the FAISS index.

We also encountered difficulties in **database design and consistency**, especially in managing relationships between assets, tags, and versions. This is being addressed by refining the schema and adding proper indexing.

On the frontend, **API integration and state management** have been challenging. We plan to improve this with better error handling and structured state management.

Lastly, **duplicate detection** is reliable for exact matches but less effective for near duplicates. We plan to enhance it by combining hashing with embedding-based similarity.

Overall, these challenges are being addressed through optimization, better design, and iterative improvements.

Tasks Pending

Tasks Pending	Team Member (to complete the task)
The Backend & System Architect needs to complete advanced APIs such as semantic search, filtering, version control, and downloads, while also strengthening RBAC, improving error handling, and optimizing overall performance and integration with other components.	Aditya Sharma
The Frontend & UI Developer needs to complete the search interface with filters, implement version history and asset preview features, improve responsiveness, and ensure smooth integration with backend APIs along with proper state management and error handling.	Ayush Bisht
The AI/ML Engineer is responsible for finalizing the semantic search using FAISS, improving embedding quality, and completing features like auto-tagging and near-duplicate detection.	Kushagra Garg
The Database & Search Engineer will focus on optimizing the PostgreSQL schema, improving relationships between assets, tags, and versions, adding indexing for faster queries, and ensuring consistency between the database and vector index.	Apratim Rana

Project Outcome / Deliverables

The main outcome of this project is a fully functional **Intelligent Digital Asset Management System (IDAMS)** that enables efficient storage, organization, and retrieval of digital assets. The system will provide a user-friendly web interface where users can upload, manage, search, and download files securely.

Key deliverables include a **React-based frontend dashboard** for interaction, a **Flask-based backend system** with RESTful APIs, and a **PostgreSQL database** for storing metadata such as asset details, tags, user information, and version history. The project also delivers an **AI-powered module** capable of generating embeddings, enabling semantic search, and supporting automatic tagging of assets.

Additionally, a **FAISS-based vector index** will be implemented for fast similarity search, along with a **duplicate detection mechanism** to reduce redundant storage. The system will also include **JWT-based authentication and role-based access control (RBAC)** for secure usage.

Other deliverables include proper **documentation**, system design details, and testing results. Overall, the project results in a scalable, intelligent platform that improves productivity, reduces manual effort, and enhances digital asset management efficiency.

Progress Overview

The project is currently **around 50% complete**, with the core foundation and initial features in place. The backend structure, basic REST APIs (authentication, upload), and database setup have been implemented. The frontend has a working interface for login, upload, and basic asset viewing. Initial work on the AI pipeline, including embedding generation and basic FAISS integration, has also been started.

In terms of progress, the **system architecture, authentication, and basic upload/storage pipeline are on track or slightly ahead of schedule**, as a working end-to-end flow from upload to storage has already been achieved.

However, some components are **slightly behind schedule**, particularly the advanced AI features such as accurate semantic search, auto-tagging, and near-duplicate detection, which require further tuning and integration. Additionally, frontend enhancements like advanced search UI, filtering, and version history display are still incomplete.

Overall, while the foundational system is stable and progressing well, the focus now is on completing intelligent features, improving accuracy, and refining the user interface to meet the final project goals.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Unit Testing	Pass	Core backend functions like authentication, upload handling, and hashing are working correctly.
Security Testing	Pass	JWT authentication and RBAC are implemented and functioning properly.
Performance Testing	Fail	System shows delays during AI processing; optimization is required.

Deliverables Progress

- **Frontend Dashboard (React UI):** In Progress
Basic interface for login, upload, and asset viewing is ready; advanced UI features are pending.
- **Backend System (Flask APIs):** In Progress
Core APIs are implemented; advanced features like full search, filtering, and versioning are still being completed.
- **Database (PostgreSQL):** In Progress
Schema is created and partially functional; optimization and relationships are being refined.
- **AI/ML Module (Tagging & Embeddings):** In Progress
Initial embedding generation is working; auto-tagging and accuracy improvements are ongoing.
- **FAISS Vector Search:** In Progress
Basic integration is done; full semantic search functionality is not yet finalized.
- **File Storage System (Local Storage):** Completed
File upload and storage are working reliably.
- **Authentication & RBAC:** In Progress
JWT authentication is implemented; RBAC needs further strengthening.
- **Duplicate Detection System:** In Progress
Exact duplicate detection works; near-duplicate detection is still being improved.
- **Documentation & Testing:** In Progress
Basic testing is done; detailed documentation and final testing are pending.

Most deliverables are **in progress**, with core functionality established and remaining work focused on enhancement, optimization, and full integration.

Codebase Information

Repository Link - <https://github.com/KushagraGarg100/Intelligent-Digital-Access-Management-System/tree/main>